The data check

Settings → Data sources → Check every data source. One button that asks every source the screener tools and qp depend on for real data, and judges whether what came back is **usable**, not just whether it answered.

Written from `src/routes/datacheck.js` (the tool's half) and `quant-platform/chart/datacheck.py` (qp's half).

Related: where each source is used (`docs/DATA-PIPELINE.md`), the keys (`docs/SETTINGS.md`).

Why it exists

A source that answers with nothing, or with numbers that are wrong, does not raise an error: the cards just come out thinner, and nobody notices for weeks. So the check asks for data that **cannot** legitimately be empty and judges the content. It is slow on purpose (up to two minutes) and runs only when pressed, never on a timer.

How to run it

- The button on the Settings tab, or
- `GET /api/datacheck?symbol=SPY` on any tool (the symbol is used for qp's bar checks; default SPY).

The tool runs its own checks and asks qp (`QP_URL`, default `http://127.0.0.1:8765`, `GET /api/datacheck`) for the rest, all at once.

Verdicts

Severity	Meaning
ok	usable, nothing to say
degraded	usable, but something is wrong enough to say: a gap, a warning, a stale file, a source missing from a set
down	not usable

The answer:

```
{ "ok": true,            // false when anything is down
  "tool": "T1", "symbol": "SPY", "ms": 41200,
  "total": 18, "passed": 15, "degraded": 2, "down": 1,
  "qpReachable": true,
  "checks": [ { "name", "ok", "severity", "detail", "fix"?, "ms"?, … } ],
  "summary": "every source returned usable data"  // or the names of what failed
}
```

Where a check knows the remedy, it carries a `fix` line.

The tool’s checks

Check	Asks	down	degraded	ok
tradingview scanner	a scan for `close > 0`, 50 rows (a filter nothing can fail)	zero rows; the first row has no usable price (the columns changed); the fetch failed	—	`N rows, first is X at $P`
news sources	every news source for AAPL	no stories from any source	stories came, but one or more sources failed; when a key is missing or rejected, `fix` names it	`N stories for AAPL`
Alpaca key	the account behind the stored key (12 s)	key rejected (401/403); could not ask	—	account number, status, `paper` or `LIVE`
industry map	the tool’s own map (`docs/MARKET-TAB.md`)	empty	under 200 symbols or under 20 industries	`N symbols across M industries`
qp (chart platform)	qp’s whole check (120 s limit)	qp not answering: then every qp check below is missing, and it says so	—	qp’s own rows are added

qp’s checks

Bar feeds

Every feed qp can load (`data_manager.LOADERS`: `alpaca`, `polygon`, `hybrid`, `hybrid_yahoo`, `yahoo`), 40 daily bars of the symbol each. One row per feed, named `feed:SYMBOL`.

down when	degraded when
the fetch failed (with a diagnosis and, for a missing key, the fix)	more than 5 % of sessions have no volume
no data returned, or no close column	—
fewer than 15 rows	—
any close is NaN, or at or below zero	—
every volume is zero (every volume test would be dead)	—
the last bar is more than 5 sessions old	—
bars out of order, or duplicated	—
a one-day move over 25 % (a split not applied, or the wrong symbol)	—

The indexes

`^GSPC` and `^IXIC` from Yahoo, judged the same way. If either fails, the row says O’Neil’s market model cannot be built without it.

Cross-source agreement

The only check that catches a feed that is **confidently wrong**. Every pair of working feeds among `yahoo`, `polygon`, `alpaca` and `hybrid_yahoo` is compared close for close.

Result	Severity
worst difference ≤ 0.5 %	ok
worst difference > 0.5 %	down
fewer than 3 shared sessions, or one side empty	degraded
fewer than 2 working feeds to compare	degraded, with the fix “configure a second feed”

RS universe

Polygon grouped-daily, the whole-market frame the RS rating, group ranks and rank-in-group need.

Result	Severity
no sessions cached	down (fix: the backfill command; needs `POLYGON_API_KEY`)
fewer than 1,000 symbols	down
fewer than 253 sessions (a year)	degraded
otherwise	ok

Files qp builds

File	Required fields	Too old after
O'Neil market model	`status`, `indexes`	36 hours
group ranks	`groups`	36 hours
industry map	`symbols`	14 days

Missing, unreadable, or missing a required field: **down**. Present but too old: **degraded** ("nothing has rebuilt it").

SEC EDGAR

The source of the C and A tables. **down** when the ticker map has fewer than 5,000 entries, when AAPL's facts give no quarters, or on a 403 (EDGAR wants a descriptive User-Agent: set `EDGAR_UA`). ok shows how many tickers are mapped and how many quarters AAPL parsed to.

Known gaps

- The news `fix` line and the Alpaca `fix` line say "Settings > API Keys", which has no inputs; keys go in `data/keys.json` (D-T4).